HW#4

Homework Report

Course	ECG 762
Student Name	Chang Wang
Student ID	2002721058
Instructor	Ebrahim Saberinia
Date	09-28-2026

Problem 01 4.1

We wish to estimate the amplitudes of exponentials in noise. The observed data are

$$x[n] = \sum_{i=1}^{p} A_i r_i^n + w[n] \; n = 0,1, \dots, N-1$$

where $w[n]$ is WGN with variance σ^2. Find the MVU estimator of the amplitudes and also their covariance. Evaluate your results for the case when $p = 2, r_1 = 1, r_2 = -1$, and N is even.

Solution:

The quantities r_i are known, and the amplitudes A_i are to be estimated. Write the observation model as

$$\mathbf{x} = \mathbf{H}\mathbf{A} + \mathbf{w}, \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N),$$

where

$$\mathbf{A} = [A_1 \quad A_2 \quad \cdots \quad A_p]^T,$$

and

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ r_1 & r_2 & \cdots & r_p \\ r_1^2 & r_2^2 & \cdots & r_p^2 \\ \vdots & \vdots & & \vdots \\ r_1^{N-1} & r_2^{N-1} & \cdots & r_p^{N-1} \end{bmatrix}.$$

Assuming that H has linearly independent columns, the MVU estimator for the Gaussian linear model is

$$\hat{\mathbf{A}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}.$$

Substituting the observation model gives

$$\hat{\mathbf{A}} = \mathbf{A} + (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{w}.$$

Therefore,

$$E[\hat{\mathbf{A}}] = \mathbf{A},$$

and

$$\mathrm{Cov}(\hat{\mathbf{A}}) = \sigma^2(\mathbf{H}^T\mathbf{H})^{-1}.$$

The estimator is unbiased and attains the CRLB, so it is MVU. For distinct real r_i, the full-column-rank condition holds when $N \geq p$.

For explicit computation,

$$[\mathbf{H}^T\mathbf{H}]_{ij} = \sum_{n=0}^{N-1} (r_i r_j)^n, [\mathbf{H}^T\mathbf{x}]_i = \sum_{n=0}^{N-1} r_i^n x[n].$$

Special case: $p = 2, r_1 = 1, r_2 = -1$, and N even

The model becomes

$$x[n] = A_1 + A_2(-1)^n + w[n].$$

The columns of **H** are the constant sequence and the alternating-sign sequence:

$$\mathbf{H} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \\ 1 & -1 \\ \vdots & \vdots \end{bmatrix}.$$

Since N is even,

$$\sum_{n=0}^{N-1}(-1)^n = 0$$

so

$$\mathbf{H}^T\mathbf{H} = \begin{bmatrix} N & 0 \\ 0 & N \end{bmatrix}.$$

Thus, the MVU amplitude estimators are

$$\hat{A}_1 = \frac{1}{N}\sum_{n=0}^{N-1} x[n],$$

$$\hat{A}_2 = \frac{1}{N}\sum_{n=0}^{N-1} (-1)^n x[n].$$

Taking the ordinary average cancels the alternating component and estimates A_1. Multiplying by $(-1)^n$ before averaging cancels the constant component and estimates A_2. Their covariance matrix is

$$\mathrm{Cov}(\hat{\mathbf{A}}) = \begin{bmatrix} \sigma^2/N & 0 \\ 0 & \sigma^2/N \end{bmatrix}.$$

Hence, both estimates have variance σ^2/N, and their covariance is zero. Since they are jointly Gaussian, they are also independent.

Problem 02 4.4

In the linear model it is desired to estimate the signal $\mathbf{s} = \mathbf{H}\boldsymbol{\theta}$. If an MVU estimator of $\boldsymbol{\theta}$ is found, then the signal may be estimated as $\hat{\mathbf{s}} = \mathbf{H}\hat{\boldsymbol{\theta}}$. What is the PDF of $\hat{\mathbf{s}}$? Apply your results to the linear model in Example 4.2 .

Solution:

The requested distribution is understood to be that of $\hat{\mathbf{s}}$. The true signal $\mathbf{s} = \mathbf{H}\boldsymbol{\theta}$ is deterministic, whereas its estimate is random because it depends on noisy observations.
For the Gaussian linear model,

$$\hat{\boldsymbol{\theta}} \sim \mathcal{N}(\boldsymbol{\theta}, \sigma^2(\mathbf{H}^T\mathbf{H})^{-1}).$$

Because

$$\hat{\mathbf{s}} = \mathbf{H}\hat{\boldsymbol{\theta}}$$

is a linear transformation of a Gaussian vector, it is also Gaussian. Its mean is

$$E[\hat{\mathbf{s}}] = \mathbf{H}E[\hat{\boldsymbol{\theta}}] = \mathbf{H}\boldsymbol{\theta} = \mathbf{s}.$$

Its covariance matrix is

$$\begin{aligned}\mathrm{Cov}(\hat{\mathbf{s}}) &= \mathbf{H}\mathrm{Cov}(\hat{\boldsymbol{\theta}})\mathbf{H}^T \\ &= \sigma^2\mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T.\end{aligned}$$

Therefore,

$$\hat{\mathbf{s}} \sim \mathcal{N}(\mathbf{s}, \sigma^2\mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T).$$

When $p < N$, this is a singular Gaussian distribution supported on the column space of $\mathbf{H}$, so it has no ordinary N-dimensional PDF. The distribution above remains valid.
Application to Example 4.2: Fourier Analysis The textbook's Example 4.2 concerns Fourier analysis.
In Example 4.2, the signal is

$$s[n] = \sum_{k=1}^{M}\left[a_k\cos\left(\frac{2\pi kn}{N}\right) + b_k\sin\left(\frac{2\pi kn}{N}\right)\right], M < \frac{N}{2}.$$

The corresponding sine and cosine columns are orthogonal, giving

$$\mathbf{H}^T\mathbf{H} = \frac{N}{2}\mathbf{I}_{2M}.$$

Substituting into the general result yields

$$\hat{\mathbf{s}} \sim \mathcal{N}\left(\mathbf{s}, \frac{2\sigma^2}{N}\mathbf{H}\mathbf{H}^T\right).$$

The covariance between two estimated signal samples is

$$\mathrm{Cov}(\hat{s}[n], \hat{s}[m]) = \frac{2\sigma^2}{N}\sum_{k=1}^{M}\cos\left(\frac{2\pi k(n-m)}{N}\right).$$

Setting $m = n$, each cosine equals 1 , so

$$\mathrm{var}(\hat{s}[n]) = \frac{2M\sigma^2}{N}.$$

Thus, each estimated sample has the marginal distribution

$$\hat{s}[n] \sim \mathcal{N}\left(s[n], \frac{2M\sigma^2}{N}\right).$$

For a fixed number of harmonics M, increasing N reduces the estimation variance.

Problem 03 4.10

If in Example 4.4 the noise samples are uncorrelated but of unequal variance or

$$\mathrm{C} = \mathrm{diag}(\sigma_0^2, \sigma_1^2, \dots, \sigma_{N-1}^2)$$

find d_n and interpret the results. What would happen to $\hat{A}$ if a single σ_n^2 were equal to zero?

Solution:

In Example 4.4, the observations are

$$x[n] = A + w[n], n = 0,1, \dots, N-1,$$

where the noise has zero mean and covariance

$$\mathbf{C} = \mathrm{diag}(\sigma_0^2, \sigma_1^2, \dots, \sigma_{N-1}^2).$$

The coefficients d_n in the textbook multiply the whitened observations $x'[n]$, rather than the original observations $x[n]$.

Initially, assume that all noise variances are positive. Choose the whitening matrix

$$\mathbf{D} = \mathrm{diag}\left(\frac{1}{\sigma_0}, \frac{1}{\sigma_1}, \dots, \frac{1}{\sigma_{N-1}}\right),$$

so that

$$\mathbf{D}^T\mathbf{D} = \mathbf{C}^{-1}, x'[n] = \frac{x[n]}{\sigma_n}.$$

From equation (4.27),

$$\hat{A} = \sum_{n=0}^{N-1} d_n x'[n], d_n = \frac{[\mathbf{D1}]_n}{\mathbf{1}^T\mathbf{D}^T\mathbf{D1}}.$$

Since

$$[\mathbf{D1}]_n = \frac{1}{\sigma_n}, \mathbf{1}^T\mathbf{D}^T\mathbf{D1} = \sum_{k=0}^{N-1}\frac{1}{\sigma_k^2},$$

the required coefficients are

$$d_n = \frac{\frac{1}{\sigma_n}}{\sum_{k=0}^{N-1}\frac{1}{\sigma_k^2}}.$$

Substituting $x'[n] = x[n]/\sigma_n$ gives

$$\hat{A} = \frac{\sum_{n=0}^{N-1} \frac{x[n]}{\sigma_n^2}}{\sum_{k=0}^{N-1} \frac{1}{\sigma_k^2}}.$$

Thus, expressed in terms of the original observations,

$$\hat{A} = \sum_{n=0}^{N-1} \alpha_n x[n], \alpha_n = \frac{\frac{1}{\sigma_n^2}}{\sum_{k=0}^{N-1} \frac{1}{\sigma_k^2}}.$$

The weights satisfy $\sum_n \alpha_n = 1$ and are inversely proportional to the noise variances. Observations with smaller noise variances receive greater weight because they are more reliable. If all noise variances are equal, the estimator reduces to the ordinary sample mean.

The variance of the estimator is

$$\text{var}(\hat{A}) = \left(\sum_{n=0}^{N-1} \frac{1}{\sigma_n^2} \right)^{-1}.$$

If a single noise variance, say σ_j^2, equals zero, then

$$E[w[j]] = 0, \text{var}(w[j]) = 0,$$

which implies $w[j] = 0$ almost surely. Therefore,

$$x[j] = A.$$

The amplitude can then be estimated exactly:

$$\hat{A} = x[j] = A, \text{var}(\hat{A}) = 0.$$

In this case, C is singular, so the inverse-covariance formula cannot be applied directly. Instead, the noiseless observation determines A without error.

Problem 04 4.13

In practice we sometimes encounter the "linear model" $\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$ but with $\mathbf{H}$ composed of random variables. Suppose we ignore this difference and use our usual estimator

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x},$$

where we assume that the particular realization of $\mathbf{H}$ is known to us. Show that if $\mathbf{H}$ and $\mathbf{w}$ are independent, the mean and covariance of $\hat{\boldsymbol{\theta}}$ are

$$E(\hat{\boldsymbol{\theta}}) = \boldsymbol{\theta}$$

$$\mathbf{C}_{\hat{\theta}} = \sigma^2 E_H[(\mathbf{H}^T\mathbf{H})^{-1}]$$

where E_H denotes the expectation with respect to the PDF of H. What happens if the independence assumption is not made?

Solution:

Consider

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w},$$

where $\boldsymbol{\theta}$ is a deterministic parameter vector and

$$E[\mathbf{w}] = \mathbf{0}, \text{Cov}(\mathbf{w}) = \sigma^2\mathbf{I}.$$

Assume that $\mathbf{H}$ has full column rank almost surely and that the required moments exist. Define

$$\mathbf{B}(\mathbf{H}) = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T.$$

Substituting the observation model into the estimator gives

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T(\mathbf{H}\boldsymbol{\theta} + \mathbf{w})$$

$$= \boldsymbol{\theta} + \mathbf{B}(\mathbf{H})\mathbf{w}.$$

Mean

Because **H** and **w** are independent,

$$E[\mathbf{w} \mid \mathbf{H}] = \mathbf{0}.$$

Conditioning on **H**, we obtain

$$E[\hat{\boldsymbol{\theta}} \mid \mathbf{H}] = \boldsymbol{\theta} + \mathbf{B}(\mathbf{H})E[\mathbf{w} \mid \mathbf{H}] = \boldsymbol{\theta}.$$

By the law of total expectation,

$$E[\hat{\boldsymbol{\theta}}] = E_{\mathbf{H}}[E[\hat{\boldsymbol{\theta}} \mid \mathbf{H}]] = \boldsymbol{\theta}.$$

Hence, the estimator is unbiased.

Covariance

Independence also implies

$$\mathrm{Cov}(\mathbf{w} \mid \mathbf{H}) = \sigma^2\mathbf{I}.$$

Therefore,

$$\begin{aligned}\mathrm{Cov}(\hat{\boldsymbol{\theta}} \mid \mathbf{H}) &= \sigma^2\mathbf{B}(\mathbf{H})\mathbf{B}(\mathbf{H})^T \\ &= \sigma^2(\mathbf{H}^T\mathbf{H})^{-1}.\end{aligned}$$

Using the law of total covariance,

$$\begin{aligned}\mathrm{Cov}(\hat{\boldsymbol{\theta}}) =& E_{\mathbf{H}}[\mathrm{Cov}(\hat{\boldsymbol{\theta}} \mid \mathbf{H})] \\ &+\mathrm{Cov}_{\mathbf{H}}(E[\hat{\boldsymbol{\theta}} \mid \mathbf{H}]).\end{aligned}$$

The second term is zero because the conditional mean is the fixed vector $\boldsymbol{\theta}$. Consequently,

$$\mathbf{C}_{\hat{\boldsymbol{\theta}}} = \sigma^2 E_{\mathbf{H}}[(\mathbf{H}^T\mathbf{H})^{-1}].$$

The inverse must be taken before the expectation; in general, these operations cannot be interchanged.

Without the independence assumption

If **H** and **w** are dependent, the unconditional condition $E[\mathbf{w}] = \mathbf{0}$ does not guarantee that

$$E[\mathbf{w} \mid \mathbf{H}] = \mathbf{0}.$$

In general,

$$E[\hat{\boldsymbol{\theta}}] = \boldsymbol{\theta} + E_{\mathbf{H}}[\mathbf{B}(\mathbf{H})E[\mathbf{w} \mid \mathbf{H}]].$$

The additional term may be nonzero, so the estimator may be biased. Moreover, the conditional noise covariance need not equal $\sigma^2\mathbf{I}$, and the stated covariance formula may also fail.

For example, consider the scalar case

$$w \sim \mathcal{N}(0, \sigma^2), H = \mathrm{sign}(w)$$

Then

$$\hat{\theta} = \frac{x}{H} = \theta + \frac{w}{H} = \theta + |w|.$$

Thus,

$$E[\hat{\theta}] = \theta + \sigma\sqrt{\frac{2}{\pi}} \neq \theta,$$

and

$$\mathrm{var}(\hat{\theta}) = \sigma^2\left(1 - \frac{2}{\pi}\right),$$

whereas the independence-based covariance formula would give σ^2 . Therefore, without independence, neither conclusion is guaranteed. Independence is sufficient but not necessary: both results still hold if

$$E[\mathbf{w} \mid \mathbf{H}] = \mathbf{0}, \mathrm{Cov}(\mathbf{w} \mid \mathbf{H}) = \sigma^2\mathbf{I}.$$